

CortexCrawler AI Production-Grade Semantic Web Extraction & Dataset Generation Platform

CortexCrawler AI is an advanced AI-powered semantic web extraction and dataset generation platform designed for creating high-quality chatbot datasets and RAG-ready knowledge bases. Unlike traditional web scrapers, the system intelligently understands webpage content, removes irrelevant information, filters meaningful knowledge, and transforms unstructured web data into structured conversational datasets.

Project Vision

The objective of the project is to build an intelligent data extraction engine capable of: Scraping websites dynamically Removing junk and irrelevant data Understanding semantic meaning Generating chatbot-ready datasets Creating instruction-tuning and conversational datasets Supporting enterprise RAG pipelines

High-Level Architecture

The system follows a multi-stage AI data processing pipeline: Frontend Input Layer API Gateway Task Queue System Distributed Crawl Scheduler Browser Crawling Engine Raw HTML Storage Content Extraction Engine Junk Removal Engine AI Relevance Filtering Semantic Chunking Engine Deduplication Engine Dataset Generation Engine Dataset Quality Validator Vector Database & Export Engine

Frontend Layer

Technology Stack: Next.js TailwindCSS shadcn/ui Responsibilities: Accept URL input Select dataset formats Display crawl progress Preview extracted data Export datasets

Backend API Layer

Technology: FastAPI Responsibilities: REST API handling Authentication management Task scheduling Dataset orchestration Communication with workers

Crawling Engine

Technology Stack: Playwright Crawl4AI Features: Dynamic webpage rendering JavaScript execution Infinite scroll handling Session persistence Async crawling Multi-page crawling

Content Extraction Engine

Technology Stack: Trafilatura Readability-lxml BeautifulSoup Responsibilities: Extract readable content Remove navigation bars Remove advertisements Extract metadata Clean raw HTML

AI Semantic Relevance Engine

Core AI Layer: BGE-M3 Embeddings GPT-5.5 Classification Responsibilities: Identify meaningful content Remove junk text Classify semantic relevance Filter low-quality content Improve dataset quality

Semantic Chunking Engine

Purpose: Split content intelligently Preserve semantic meaning Create chatbot-friendly chunks Optimize retrieval performance Methods: Heading-aware chunking Embedding-aware splitting Semantic boundary detection

Deduplication Engine

Technology Stack: FAISS MinHash Cosine Similarity Responsibilities: Remove duplicate content Detect semantic similarity Prevent repetitive training data

Dataset Generation Engine

Supported Formats: Question-Answer datasets Instruction tuning datasets ChatML conversational datasets RAG-ready document chunks Example Output: { "messages": [{"role": "user", "content": "What is AI?"}, {"role": "assistant", "content": "AI refers to..." }] }

Dataset Quality Validation

Metrics: Semantic relevance score Readability score Duplication score Toxicity score Factual density Hallucination probability

Storage Architecture

Databases Used: MongoDB for raw HTML storage PostgreSQL for structured metadata Qdrant for vector embeddings

Task Queue & Distributed Processing

Technology Stack: Redis Celery Responsibilities: Distributed task execution Async processing Queue management Worker orchestration

Deployment Architecture

Technology Stack: Docker AWS Cloudflare Grafana Prometheus Features: Containerized deployment Scalable infrastructure Performance monitoring System observability

Folder Structure

project-root/  frontend/  backend/   api/   crawler/   extraction/   filtering/   embeddings/   chunking/   deduplication/   dataset_generation/   validation/ exports/ workers/ models/ datasets/ docker/ scripts/ tests/

Project Roadmap

Phase 1: Single-page scraping Content cleaning JSONL export Phase 2: AI relevance filtering Semantic chunking QA generation Phase 3: Multi-page crawling Deduplication Vector database integration Phase 4: Autonomous crawling agents Enterprise RAG support Distributed crawling infrastructure

Final Positioning

CortexCrawler AI is not a traditional web scraper. It is a semantic knowledge extraction and AI dataset generation platform designed for enterprise-grade chatbot training, LLM fine-tuning, and retrieval-augmented generation systems. The platform combines AI engineering, NLP, semantic search, distributed crawling, data engineering, and modern LLM pipelines into a unified intelligent extraction architecture.
